

Convergence of iterated SOCP refinement for CAZAC feasibility

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Abstract

Constant-amplitude zero-autocorrelation (CAZAC) sequences have constant magnitude in time and flat Fourier magnitude. We study an iterative construction that relaxes these two nonconvex equality constraints to convex inequalities and repeatedly solves a second-order cone program, using the previous optimizer as the next objective direction. At relaxation parameter $\tau = 1$, the CAZAC systems are exactly the maximal-norm points of the resulting compact convex set.

For each fixed $\tau \geq 1$, every nonzero trajectory converges to a single fixed point and has finite total path length. This conclusion follows by applying a known Kurdyka–Łojasiewicz/Frank–Wolfe descent template to the CAZAC norm-maximization formulation. Global convergence is not global success: we construct exposed non-CAZAC fixed points, so no deterministic choice of optimizer can guarantee convergence to the CAZAC set. Locally, a regularity hypothesis gives a sharper conclusion. Near a regular CAZAC limit the constraint slack grows linearly, the relevant KL exponent is zero, and the iteration reaches the limit exactly after finitely many steps.

We compute the regularity condition at every Zadoff–Chu tuple. If $n = \prod_p p^{a_p}$ and $d_{\max} = \prod_p p^{\lfloor a_p/2 \rfloor}$, then the active-gradient dependency space has dimension $d_{\max}$, for odd and even n alike. Thus the local finite-arrival theorem applies at Zadoff–Chu limits exactly when n is squarefree. Earlier constructions already supplied continuous CAZAC families at non-squarefree lengths; the result here is the exact local count and its connection to convergence. We also apply the general convergence argument to the elliptope and prove a partial symmetry-restricted statement for Björck sequences; full Björck regularity is prior work. The paper separates unconditional proofs, the regularity-conditional result, numerical evidence, and open basin-of-attraction questions.

1 Introduction

A constant-amplitude zero-autocorrelation (CAZAC) sequence is a finite complex sequence whose entries all have the same magnitude and whose discrete Fourier transform also has constant magnitude. Such sequences are useful in waveform design, synchronization, and coding, and they form a nonconvex feasibility set: one must satisfy a torus constraint in time and another in frequency at the same time. Zadoff–Chu chirps give explicit examples at every length [8], but the broader geometry of the feasible set and the behavior of iterative construction methods remain difficult.

This paper studies a convex-relaxation method for that feasibility problem. For N coupled length- n sequences, let C_τ impose upper bounds on each time-domain amplitude and on the coupled Fourier energy at each frequency. The set C_τ is compact, convex, semialgebraic, and second-order-cone representable. At $\tau = 1$, its points of largest squared norm are exactly the coupled CAZAC systems. Starting from a nonzero vector, we repeatedly maximize the linear functional defined by

the current iterate:

$$x^{\ell+1} \in \arg \max_{x \in C_\tau} \langle x^\ell, x \rangle_{\mathbb{R}}.$$

This is iterated linear optimization, or a generalized power method, and each step is an SOCP.

The first result separates convergence of the iteration from success of the construction. Theorem 4.1 proves that every measurable optimizer selection has finite total path length and converges to one fixed point. The descent mechanism is not new: it is the quadratic-objective instance of the Kurdyka–Łojasiewicz/Frank–Wolfe framework developed in greater generality by Aktaş and Kroer [4], building on standard KL descent theory [1]. The CAZAC-specific contribution is the norm-maximization reduction that makes this framework apply. Theorem 5.1 then shows why convergence alone cannot guarantee success: exposed non-CAZAC fixed points exist, so a deterministic selection rule cannot guarantee arrival at the target set.

The local theory is stronger. Under a regularity condition on the active constraint gradients, Theorem 6.4 proves linear growth of the constraint slack, KL exponent zero, and exact arrival at the limiting CAZAC point after finitely many steps. Theorem 7.1 determines when this hypothesis holds at Zadoff–Chu tuples. If $n = \prod_p p^{a_p}$, the dependency space has dimension $d_{\max} = \prod_p p^{\lfloor a_p/2 \rfloor}$; hence regularity holds exactly for squarefree n . The proof uses closure of quadratic chirps under the DFT to replace a rank computation by the congruence $t^2 \equiv 0 \pmod{n}$.

Earlier CAZAC work already shows why degeneracy can occur. Backelin [10], Popović [11], and Björck–Saffari [13] already constructed continuous families at non-squarefree lengths and therefore anticipated the existence of the degeneracy. Theorem 7.1 supplies the exact count at every Zadoff–Chu tuple, for both parities, and connects it to the convergence rate. The paper does not identify its dependency space with those earlier families, and it does not claim that the existence of non-squarefree degeneracy is new.

The same method also applies to the ellipsope. The global finite-length argument transfers to the ellipsope of correlation matrices, where it proves single-point convergence even though the fixed-point set is positive-dimensional for $n > 3$. The pointwise attractive-fixed-point dichotomy follows from Felzenszwalb, Klivans, and Paul [3], but it remains open whether almost every initialization reaches a rank-one point. For Björck’s Legendre-symbol construction, we prove regularity only on a symmetry-invariant subspace; full regularity is proved by Benoist [14] and is not a contribution of this paper.

For $N = 1$ and odd prime n , Benoist [14, Prop. A.2] proves projective transversality at Gaussian chirps, the prime-length counterpart of hypothesis (R). Theorem 7.1 supplies the exact dependency count and extends the calculation to every length, both parities, every coprime root, and coupled systems with $N \geq 1$.

The next section defines the relaxation and its fixed points. Theorem A and the ellipsope application follow, then the obstruction theorem, the regularity-conditional local result, and the Zadoff–Chu dependency calculation. The paper closes with open problems and a claim-by-claim verification ledger. Numerical experiments support selected calculations and basin questions, but they are kept distinct from the proofs.

2 Setup

Fix $n \geq 1$ and $N \geq 1$. Let F denote the unnormalized discrete Fourier transform on $\mathbb{C}^n$, $F(u)(k) = \sum_m u(m) e^{-2\pi i k m / n}$, so Parseval’s identity reads $\|Fu\|^2 = n\|u\|^2$. The state space is $x = (x_0, \dots, x_{N-1}) \in (\mathbb{C}^n)^N \cong \mathbb{R}^{2Nn}$, with real inner product $\langle u, v \rangle_{\mathbb{R}} = \operatorname{Re} \langle u, v \rangle$ and induced norm $\|x\|^2 = \sum_{j,k} |x_j(k)|^2$. Write the *coupled spectrum* $S_x(k) = \sum_{j=0}^{N-1} |F(x_j)(k)|^2$.

Definition 2.1 (Coupled CAZAC system). x is a coupled CAZAC system if $|x_j(k)| = 1$ for all j, k and $S_x(k) = Nn$ for all k . Write $\text{CAZ}(n, N)$ for this set; for $N = 1$ this is the classical CAZAC condition. $\text{CAZ}(n, N)$ is nonempty for every $n \geq 1, N \geq 1$: a Zadoff–Chu sequence exists for every length n [8], and N copies of one satisfy the coupled constraint.

For $\tau \geq 1$, the convex relaxation is

$$C_\tau = \{x \in (\mathbb{C}^n)^N : |x_j(k)| \leq \tau \forall j, k, \quad S_x(k) \leq Nn \forall k\}.$$

C_τ is compact (amplitude bounds), convex, semialgebraic, and second-order-cone representable (each constraint is a Euclidean-norm bound on a linear image of x); it is invariant under global phase rotation of each x_j , cyclic shifts, and (for the coupled constraint) any transformation preserving S_x .

Iteration. Given $x^0 \neq 0$, the algorithm is *iterated linear optimization*:

$$x^{\ell+1} \in \arg \max_{x \in C_\tau} \langle x^\ell, x \rangle_{\mathbb{R}}. \quad (\star)$$

The argmax may be non-unique; $(\star)$ permits any measurable selection. The reference implementation performs ten preliminary solves with $\tau = 1/\ell$ for $\ell = 1, \dots, 10$, then resets to $\tau = 1$ for the main phase. All theorems below concern an iteration at one fixed $\tau \geq 1$ (in particular $\tau = 1$), and hold for any measurable selection rule. The preliminary solves merely choose the input direction for the analyzed main phase; they are not part of any theorem.

Definition 2.2 (Fixed point). x^* is a fixed point of $(\star)$ if $x^* \in \arg \max_{x \in C_\tau} \langle x^*, x \rangle_{\mathbb{R}}$, equivalently $\sigma_{C_\tau}(x^*) = \|x^*\|^2$, where $\sigma_C(r) = \max_{x \in C} \langle r, x \rangle_{\mathbb{R}}$ is the support function of C .

Three increasingly strong convergence questions organize everything that follows: (1) does x^ℓ converge to a single point, not merely have limit points; (2) is that limit CAZAC; (3) at what rate.

3 The iteration as a generalized power method

Proposition 3.1. Let $f(x) = \frac{1}{2}\|x\|^2$, so f is convex and $\nabla f(x) = x$. Then $(\star)$ is exactly

$$x^{\ell+1} \in \arg \max_{x \in C_\tau} \langle \nabla f(x^\ell), x \rangle_{\mathbb{R}}.$$

This is the conditional-gradient or generalized-power-method update for maximizing f over the compact convex set C_τ [5], also called iterated linear optimization [3].

Proof. Immediate from the definitions. □

Proposition 3.2 (Max-norm points are exactly CAZAC points). For every $x \in C_\tau$ ($\tau \geq 1$), $\|x\|^2 \leq Nn$. For $\tau = 1$, equality holds iff $x \in \text{CAZ}(n, N)$. The maximum is attained for every n, N, τ .

Proof. By Parseval, $\|x\|^2 = \sum_j \|x_j\|^2 = \frac{1}{n} \sum_k S_x(k) \leq \frac{1}{n} \cdot n \cdot Nn = Nn$, with equality iff $S_x(k) = Nn$ for every k . Separately, at $\tau = 1$, $\|x\|^2 = \sum_{j,k} |x_j(k)|^2 \leq Nn$ with equality iff every $|x_j(k)| = 1$. Both chains tight simultaneously forces both CAZAC conditions; conversely a CAZAC system attains Nn . Attainment: N copies of a Zadoff–Chu sequence of length n lie in $C_1 \subseteq C_\tau$ with $\text{norm}^2 = Nn$. □

This reframes the nonconvex feasibility problem “find a CAZAC system” as the convex maximization $\max_{x \in C_1} \|x\|^2$, whose optimal value Nn is known a priori: an iterate is δ -suboptimal iff $Nn - \|x^\ell\|^2 = \delta$.

Proposition 3.3 (Monotone ascent, square-summable increments). *Along the iteration at fixed τ , with $x^\ell \neq 0$ and $x^0 \in C_\tau$:*

- (1) $\langle x^\ell, x^{\ell+1} \rangle_{\mathbb{R}} \geq \|x^\ell\|^2$;
- (2) $\|x^{\ell+1}\| \geq \|x^\ell\|$, and $\|x^\ell\| \uparrow \rho \leq \sqrt{Nn}$;
- (3) $\|x^{\ell+1} - x^\ell\|^2 \leq \|x^{\ell+1}\|^2 - \|x^\ell\|^2$, hence $\sum_{\ell \geq 0} \|x^{\ell+1} - x^\ell\|^2 \leq Nn - \|x^0\|^2$ and $\|x^{\ell+1} - x^\ell\| \rightarrow 0$.

Proof. Parts (1)–(2) and the bound in (3) instantiate the argument of [3]’s Theorem 1 for this C_τ ; only the explicit telescoped constant $Nn - \|x^0\|^2$ in (3) is not in their statement, since it uses Prop. 3.2’s CAZAC-specific bound Nn in place of a general compactness bound. (1) $x^\ell \in C_\tau$ is feasible for the SOCP whose optimizer is $x^{\ell+1}$, so the optimum is at least $\langle x^\ell, x^\ell \rangle_{\mathbb{R}} = \|x^\ell\|^2$; this is where $x^\ell \in C_\tau$ is used, which is why the hypothesis is stated for x^0 and not just $x^{\ell \geq 1}$ (every $x^{\ell+1}$ for $\ell \geq 0$ is automatically in C_τ , being an argmax over C_τ). (2) Cauchy–Schwarz: $\|x^\ell\| \|x^{\ell+1}\| \geq \langle x^\ell, x^{\ell+1} \rangle_{\mathbb{R}} \geq \|x^\ell\|^2$; divide by $\|x^\ell\|$; monotone and bounded above by Prop. 3.2. (3) $\|x^{\ell+1} - x^\ell\|^2 = \|x^{\ell+1}\|^2 + \|x^\ell\|^2 - 2\langle x^\ell, x^{\ell+1} \rangle_{\mathbb{R}} \leq \|x^{\ell+1}\|^2 - \|x^\ell\|^2$ by (1); telescope and apply (2). $\square$

The hypothesis $x^0 \in C_\tau$ costs nothing. Whatever input direction is handed to a fixed- τ phase, its first optimizer lies in C_τ by construction. Re-indexing that phase from its first optimizer therefore restores the hypothesis with no loss; every result below that depends on Prop. 3.3 inherits this same one-step re-indexing.

Square-summable increments alone do not imply convergence of the full sequence (harmonic-type sums can wander); closing exactly this gap is Theorem A.

Proposition 3.4 (Limit points are fixed points; CAZAC points are fixed points). *Every limit point of (x^ℓ) is a fixed point of $(\star)$. Every $x^* \in \text{CAZ}(n, N)$ is a fixed point of the $\tau = 1$ iteration, and moreover a global maximizer of its own functional: $\arg \max_{x \in C_1} \langle x^*, x \rangle_{\mathbb{R}} \ni x^*$ with optimal value Nn .*

Proof. Let $x^{\ell_j} \rightarrow \bar{x}$. By Prop. 3.3(3), $x^{\ell_j+1} \rightarrow \bar{x}$ as well. σ_{C_τ} is continuous (finite convex on $\mathbb{R}^{2Nn}$), and $\langle x^{\ell_j}, x^{\ell_j+1} \rangle_{\mathbb{R}} = \sigma_{C_\tau}(x^{\ell_j})$ by definition of the update; pass to the limit. For the second claim, any $y \in C_1$ satisfies $\langle x^*, y \rangle_{\mathbb{R}} \leq \|x^*\| \|y\| \leq \sqrt{Nn} \sqrt{Nn} = Nn = \|x^*\|^2$ by Cauchy–Schwarz and Prop. 3.2, with equality at $y = x^*$. $\square$

The same Cauchy–Schwarz argument shows any maximal-norm point of *any* compact convex set is a fixed point of iterated linear optimization; nothing CAZAC-specific is used beyond Prop. 3.2 itself — a fact that drives the ellipsope corollary in §4.1.

4 Theorem A — global convergence

Theorem 4.1 (Theorem A — Proved, unconditional). *Fix $\tau \geq 1$. For any $x^0 \neq 0$ and any measurable selection $x^{\ell+1} \in \arg \max_{x \in C_\tau} \langle x^\ell, x \rangle_{\mathbb{R}}$, the sequence (x^ℓ) has finite length, $\sum_\ell \|x^{\ell+1} - x^\ell\| < \infty$, and converges to a single fixed point $\bar{x} \in C_\tau$.*

Proof. Apply the abstract descent theorem of Attouch, Bolte, and Svaiter [1, Thm. 2.9] to $F(x) = -\frac{1}{2}\|x\|^2 + \iota_{C_\tau}$ (so minimizing F is maximizing $\|x\|^2/2$ on C_τ):

- **Sufficient decrease** ($a = \frac{1}{2}$): exactly Prop. 3.3(3).

- **Relative error** ($b = 1$): first-order optimality of the argmax step gives $x^\ell \in N_{C_\tau}(x^{\ell+1})$, the normal cone at the new iterate. Since $\partial F(x^{\ell+1}) = -x^{\ell+1} + N_{C_\tau}(x^{\ell+1})$ (sum rule for C^1 plus indicator of a convex set), the vector $w = x^\ell - x^{\ell+1}$ lies in $\partial F(x^{\ell+1})$, and $\|w\| = \|x^{\ell+1} - x^\ell\|$ *exactly*: the subgradient residual is literally the step. This identity is specific to $f = \frac{1}{2}\|x\|^2$; for a generic convex objective the relative-error condition would be a genuine additional hypothesis, but here it is free. (The identical construction, for a general smooth strongly convex objective where it holds only as a Lipschitz-constant inequality rather than an exact equality, is Aktas–Kroer’s [4] Lemma 3.1 eq. (27)–(28); our case is the special instance where that Lipschitz constant is exactly 1, so the inequality becomes an equality — this observation is a corollary of their general lemma, not an independent discovery.)
- **Compactness/continuity**: iterates lie in compact C_τ ; F is continuous there.
- **KL property**: C_τ is defined by polynomial (in)equalities and F is polynomial plus the indicator of a semialgebraic set, hence semialgebraic, hence KL [2].

Critical points of F are exactly the fixed points of $(\star)$: $0 \in \partial F(x^*)$ iff $x^* \in N_{C_\tau}(x^*)$ (the sum rule above) iff x^* maximizes $\langle x^*, \cdot \rangle_{\mathbb{R}}$ over C_τ , which is the Fixed-point Definition of Section 2. The theorem then gives finite length and convergence to a single critical point. $\square$

The theorem is silent on *which* fixed point the limit is; that is exactly where Theorem B enters. Selection-independence, and the fact that the presolve is finitely many extra solves, mean the conclusion applies verbatim to any concrete implementation of $(\star)$.

Remark 4.2 (Novelty check against existing CAZAC-construction algorithms). The closest prior iterative CAZAC-construction method is IPUC (Iterative Projection onto Unit Circle) [9]: alternate projecting onto the unit-modulus constraint in time and in frequency, from a random seed. IPUC’s own authors report, at length $n = 50$, 8 of 20 trajectories with discrepancy frozen above 10^{-3} after 10^5 iterations, and state plainly that “the analysis of this iterative function is beyond the scope of this paper”; their practical remedy is to restart from a new random seed rather than continue iterating. Theorem A gives what IPUC’s own presentation explicitly does not attempt: an unconditional proof that (a different, convex-relaxation-based) iteration converges, from every initialization, to a single fixed point, with finite total path length. The two algorithms are not the same iteration — IPUC projects directly onto the nonconvex unimodular constraint in both domains, while $(\star)$ relaxes to the convex body C_τ — so this is not a convergence proof for IPUC itself, but it is, as far as we have found, the first provable convergence guarantee for an iterative CAZAC-construction method in this literature.

4.1 A second worked example: the elliptope

The proof of Theorem 4.1 uses nothing about C_τ beyond compactness and convexity, except at the KL step, which needs semialgebraicity. Stated in general: for *any* compact convex $C \subset \mathbb{R}^d$ and the iteration $x^{\ell+1} \in \arg \max_{x \in C} \langle x^\ell, x \rangle$, sufficient decrease (Prop. 3.3(3)’s proof uses only feasibility of x^ℓ for the next problem and Cauchy–Schwarz) and the relative-error identity (the normal-cone argument above, using only $x^\ell \in N_C(x^{\ell+1})$ and $\nabla(\frac{1}{2}\|x\|^2) = x$) hold unconditionally; compactness gives continuity. If C is also semialgebraic, KL holds [2] and Theorem A’s conclusion — convergence to a single fixed point, not merely a connected limit set — follows for that C with no CAZAC-specific input.

We exercise this on a genuinely different C : the **elliptope**

$$\mathcal{E}_n = \{X \in \mathbb{R}^{n \times n} : X = X^\top, X \succeq 0, X_{ii} = 1 \ \forall i\},$$

the set of $n \times n$ correlation matrices, with iteration $X^{\ell+1} \in \arg \max_{X \in \mathcal{E}_n} \langle X^\ell, X \rangle$ ($\langle A, B \rangle = \text{tr}(AB)$). This is Felzenszwalb, Klivans, and Paul’s own example [3]: a case where they concede the fixed-point set is a continuum for $n > 3$, and where their own worked instance stops at set-level convergence.

Proposition 4.3 (Proved). $\mathcal{E}_n$ is compact, convex, and semialgebraic.

Proof. Convexity of the constraints is immediate. Compactness: $X \succeq 0$ with $X_{ii} = 1$ gives $|X_{ij}| \leq \sqrt{X_{ii}X_{jj}} = 1$ by Cauchy–Schwarz on the associated Gram vectors, so $\mathcal{E}_n$ is bounded, and closed as an intersection of closed sets, hence compact. Semialgebraicity: for symmetric X , $X \succeq 0$ is equivalent by Sylvester’s criterion to the finite conjunction of polynomial inequalities “every principal minor of X is ≥ 0 ”; the diagonal constraints are polynomial equalities. So $\mathcal{E}_n$ is a basic semialgebraic set (a spectrahedron). $\square$

Proposition 4.4 (Proved). *Sufficient decrease, relative error, and compactness/continuity transfer verbatim from Prop. 3.3(3) and the normal-cone argument of Theorem 4.1, replacing “ C_τ ” by “ $\mathcal{E}_n$ ” and $\langle \cdot, \cdot \rangle_{\mathbb{R}}$ by $\text{tr}(\cdot \cdot)$.*

Proof. Neither source proof uses any CAZAC structure; both use only convexity, Cauchy–Schwarz, and the normal-cone characterization of the argmax, all of which hold verbatim on $\mathcal{E}_n \subset \mathbb{R}^{\binom{n+1}{2}}$. $\square$

Proposition 4.5 (Proved, $n > 3$). *For $n > 3$, let*

$$\mathcal{F} = \{X \in \mathcal{E}_n : X \in \arg \max_{Y \in \mathcal{E}_n} \langle X, Y \rangle\}$$

be the fixed-point set of the elliptope iteration. It is infinite, hence (being semialgebraic) a genuine positive-dimensional continuum, not a singleton or a finite set. The restriction $n > 3$ is necessary: at $n = 2$, $\mathcal{E}_n = \{[\begin{smallmatrix} 1 & r \\ r & 1 \end{smallmatrix}]\} : |r| \leq 1\}$ and one checks directly that $\mathcal{F} = \{r \in \{-1, 0, 1\}\}$ is finite, so the statement is false without $n > 3$.

Proof. $\mathcal{F}$ is semialgebraic: $\mathcal{E}_n$ is semialgebraic (Prop. 4.3) and $X \mapsto \sigma_{\mathcal{E}_n}(X) := \max_{Y \in \mathcal{E}_n} \langle X, Y \rangle$ is semialgebraic by the Tarski–Seidenberg theorem (a projection of a semialgebraic set), so $\mathcal{F} = \{X \in \mathcal{E}_n : \langle X, X \rangle = \sigma_{\mathcal{E}_n}(X)\}$ is cut out by polynomial (in)equalities. A semialgebraic subset of $\mathbb{R}^{\binom{n+1}{2}}$ that is infinite cannot be 0-dimensional (an o-minimal set with finitely many points in every cell is finite), so it suffices to exhibit infinitely many fixed points. Felzenszwalb, Klivans, and Paul [3] state exactly this: $\mathcal{L}_n$ (their name for $\mathcal{E}_n$) has infinitely many fixed points for $n > 3$ — this, not the example below, is where the hypothesis $n > 3$ enters and cannot be dropped. Concretely, $X^* = I_n$ is one such fixed point (for any n), and its optimal face is degenerate in the strongest possible sense: $\text{tr}(I_n X) = \sum_i X_{ii} = n$ identically on $\mathcal{E}_n$, so the linear functional $\langle I_n, \cdot \rangle$ is constant there and $\arg \max_{X \in \mathcal{E}_n} \langle I_n, X \rangle = \mathcal{E}_n$ itself — every feasible point, including every rank-1 correlation matrix $vv^\top$, is simultaneously optimal for I_n ’s own direction, which in particular makes I_n a fixed point with a machine-exact identity. This illustrates the phenomenon but is not itself the infinitude argument: membership of $vv^\top$ or other points in $\arg \max \langle I_n, \cdot \rangle$ does not make them fixed points of the map, since a fixed point must be optimal for *its own* direction, not I_n ’s; the infinitude of $\mathcal{F}$ rests on [3]’s cited count, not on this example alone. $\square$

Proposition 4.6 (Verified numerically, $n = 5$). *Running the iteration on $\mathcal{E}_5$ from four starting points — $X^0 = I_5$; I_5 plus a small random perturbation renormalized to unit diagonal; and two independent random correlation matrices — using an interior-point SDP solver (CLARABEL via cvxpy), all four trajectories show $\|X^{\ell+1} - X^\ell\|_F \rightarrow 0$ (to $\sim 10^{-15}$ by iteration 15) and settle on a single*

matrix. Three of the four escape the fixed-point continuum entirely and converge to a rank-1 correlation matrix $vv^\top$ (eigenvalues $(0,0,0,0,5)$ to machine precision); the trajectory started at $X^0 = I_5$ is an edge case in which the solver’s own deterministic tie-break returns I_5 again, a valid fixed point, not a counterexample. Script: `experiments/verify_elliptope_convergence.py` in the companion repository [16]. A broader sweep (same repository, `experiments/verify_elliptope_rank_sweep.py`) of 140 random and Gaussian-perturbed trials across $n \in \{3, 4, 5, 8, 10, 15, 20\}$ finds **every trajectory converges to rank 1**, no exceptions. This remains verified numerically only, at these tested n and trial counts, not a proof for every n or every trajectory.

Theorem 4.7 (Elliptope convergence — tiers as stated). *Because $\mathcal{E}_n$ is compact, convex, and semialgebraic (Prop. 4.3), Theorem A’s general form applies to it directly, for every n : the iterated-linear-optimization sequence on the elliptope converges to a single fixed point for every initialization — **Proved**. For $n > 3$ this holds **despite the fixed-point set being an infinite continuum** (Prop. 4.5; for $n \leq 3$ the fixed-point set is finite, so the continuum caveat is vacuous there but convergence is still unconditional, by the same general argument). Which fixed point is reached is settled only partially: the vertices (rank-1 points) are exactly the fixed points whose full neighborhood converges to them, and no other fixed point has this property, **by direct citation** to [3]’s own Theorem 20 (Remark 4.9 below). That generic nearby trajectories escape the continuum and settle specifically onto a rank-1 point is **verified numerically** (140 trials, Prop. 4.6), not established for every n or every trajectory. Whether the measure of initializations converging elsewhere is exactly zero is **open** (Conjecture, filed as Q7 below).*

Remark 4.8 (Relation to Felzenszwalb–Klivans–Paul). FKP’s Theorem 1 (limit points exist, are fixed points, form a connected set) is the same starting point as Prop. 3.4 here. From an *arbitrary* initialization, on a fixed-point set that is a positive-dimensional continuum for $n > 3$ (Prop. 4.5), their only route to single-point convergence is an unnumbered remark that a connected *finite* fixed-point set is a singleton, which is structurally inapplicable here; a full read of the proof of their Theorem 1 confirms it only ever derives connectedness and never separately invokes finiteness. (FKP’s own Theorem 20 does give single-point convergence from a *restricted* class of initializations — any point within distance 1 of a vertex converges to it in one step, per Remark 4.9 below — but this does not cover an arbitrary starting point, which is the setting Theorem 4.7 needs.) Theorem 4.7 closes exactly this gap via the KL/finite-length route [1] rather than the connectedness-plus-finiteness route, and is therefore not subsumed by [3] even in the positive-dimensional-face case. This route itself, as noted at Theorem A, is not unique to this paper: Aktaş–Kroer’s [4] KL/Frank-Wolfe framework (built on a different but interchangeable abstract descent theorem, Bolte–Sabach–Teboulle–Vaisbourd) directly delivers the same elliptope conclusion — every hypothesis of their Theorem 3.2 (strongly convex smooth objective, nonempty compact constraint set, KL) is verified here by Prop. 4.3, so this is a second, independent route to Theorem 4.7, not merely a plausible one. [4] does not name FKP, does not run its framework on the elliptope itself under that name (its own Max-Cut application factorizes to a product of spheres via Burer–Monteiro, §4.3), and does not discuss positive-dimensional critical sets; the comparison to FKP should be read as “a route FKP’s own paper does not take,” not as “a route no other paper could have taken.”

Remark 4.9 (FKP’s own Theorem 20 bears directly on Q7). Although [3] cannot prove Theorem 4.7’s single-point convergence claim, its own Theorem 20 answers a large part of Q7 (below) by direct citation, without any new argument here: *the vertices of $\mathcal{E}_n$ (the rank-1 sign matrices $vv^\top$) are exactly the **attractive** fixed points of the map $X \mapsto \arg \max_{Y \in \mathcal{E}_n} \langle X, Y \rangle$* , in the pointwise sense that an entire neighborhood of a vertex converges to it, while every non-vertex fixed point admits (their Prop. 19) an explicit curve of nearby points whose norm strictly increases, so it cannot attract a full

neighborhood either. This is a genuine dichotomy, not a numerical observation, and it is stronger than what either paper’s abstract convergence machinery gives alone: it says which fixed points *can* be limits of nearby trajectories, complementing (not overlapping) Theorem 4.7’s unconditional guarantee that *some* single limit exists. It falls short of Q7 as literally stated for a precise reason: [3]’s Prop. 19 produces, in every neighborhood of a non-vertex fixed point, *at least one* nearby point that escapes it — exactly what “not attractive” (their Def. 6) requires — but this does not show *every* nearby point escapes, which is what “repelling” (their Def. 7) would additionally require. Whether non-vertex elliptope fixed points are repelling outright, or can be neither attractive nor repelling (as [3]’s own §3, Example 5, shows can happen in general iterated-linear-optimization systems, for an unrelated three-dimensional cone, not the elliptope itself) is not settled by Theorem 20 either way. If some non-vertex elliptope point is neither, Theorem 20 alone would not rule out a positive-measure set of initializations converging to it through a collective, not individually attractive, mechanism. Q7’s residual open content is exactly this measure-zero refinement, structurally the same gap as Q6.

5 Theorem B — obstruction: the limit need not be CAZAC

Theorem 5.1 (Theorem B — Proved, unconditional). (a) *The fixed points of $(\star)$ at $\tau = 1$ with every amplitude constraint slack are exactly the points whose coupled spectral profile satisfies $S_x(k) = Nn$ on some proper subset $K \subsetneq \{0, \dots, n-1\}$ of frequencies and $S_x(k) = 0$ off K (i.e. $\hat{x}_j(k) = 0$ for every j , when $k \notin K$), with time-domain modulus strictly less than 1 everywhere; such a point has $\|x\|^2 = N|K|$. For $N = 1$ this forces $|\hat{x}(k)| = \sqrt{n}$ exactly on K ; for $N \geq 2$ only the coupled sum is pinned, and the individual $|\hat{x}_j(k)|$ are otherwise free subject to $\sum_j |\hat{x}_j(k)|^2 = Nn$ for $k \in K$.*

(b) *There exist non-CAZAC fixed points that are the **unique** maximizer of their own linear functional over C_1 — exposed points. For $n = 2$: $x^* = (1, t)$, $t = \sqrt{2} - 1$, is such a point; for $N = 2$, its diagonal doubling $((1, t), (1, t))$ is such a point.*

(c) *Consequently no deterministic convergence-to-CAZAC theorem is possible: every measurable selection rule in $(\star)$ retains the point of (b) exactly, once reached.*

Proof of (a), general N . By Prop. 3.1, $x^* \neq 0$ is a fixed point iff it is a KKT point of $\max_y \langle x^*, y \rangle_{\mathbb{R}}$ subject to $g_{j,m}(y) = |y_j(m)|^2 - 1 \leq 0$ (each block $j = 1, \dots, N$) and the single *shared* constraint $h_k(y) = S_y(k) - Nn \leq 0$; this is a convex program and Slater’s condition holds at $y = 0$, so KKT is necessary and sufficient. The real gradients are $\nabla_{y_j} g_{j,m}(y) = 2y_j(m)e_m$ and, since $S_y(k) = \sum_j |\hat{y}_j(k)|^2$ couples all blocks, $\nabla_{y_j} h_k(y) = 2F^* E_k F y_j$ for every j with the *same* multiplier μ_k — this coupling is the source of the difference from $N = 1$. KKT at x^* reads, per block, $x_j^* = \Lambda_j x_j^* + F^* M F x_j^*$ for diagonal $\Lambda_j, M \geq 0$ complementary to the active sets, with M shared across j . If every amplitude constraint is slack then each $\Lambda_j = 0$, so $x_j^* = F^* M F x_j^*$ for every j ; applying F and using $F F^* = nI$ gives $\hat{x}_j^* = n M \hat{x}_j^*$ coordinatewise for every j , i.e. for each k either $\hat{x}_j^*(k) = 0$ for every j , or $\mu_k = 1/n$. Let $K = \{k : \exists j, \hat{x}_j^*(k) \neq 0\} \neq \emptyset$; for $k \in K$, complementary slackness on the *shared* constraint h_k with $\mu_k = 1/n > 0$ forces the active condition $S_x^*(k) = Nn$ — the coupled sum, not each block’s own modulus, since h_k is one constraint shared by all N blocks. Parseval (summed over blocks) gives $\|x^*\|^2 = \frac{1}{n} \sum_k S_x^*(k) = \frac{1}{n} |K| \cdot Nn = N|K|$; if $|K| = n$ this forces $\|x^*\|^2 = Nn$, hence (Prop. 3.2) $x^* \in \text{CAZ}(n, N)$, contradicting amplitude-slackness. So $K \subsetneq \{0, \dots, n-1\}$. For $N = 1$, $S_x(k) = |\hat{x}(k)|^2$, so the coupled condition $S_x(k) = n$ collapses to the per-block statement $|\hat{x}(k)| = \sqrt{n}$; for $N \geq 2$ the individual $|\hat{x}_j(k)|$ are unconstrained beyond $\sum_j |\hat{x}_j(k)|^2 = Nn$ (e.g. $N = 2, n = 2$: $\hat{x}_0(0) = \sqrt{3}, \hat{x}_1(0) = 1$ satisfies $S_x(0) = 4 = Nn$ with neither modulus equal to

$\sqrt{n} = \sqrt{2}$). Conversely, for such x^* and any $y \in C_1$, Cauchy–Schwarz per frequency gives $\langle x^*, y \rangle_{\mathbb{R}} = \frac{1}{n} \sum_{k \in K} \sum_j \operatorname{Re}(\widehat{x}_j^*(k) \widehat{y}_j(k)) \leq \frac{1}{n} \sum_{k \in K} \sqrt{S_{x^*}(k)} \sqrt{S_y(k)} \leq \frac{1}{n} \sum_{k \in K} \sqrt{Nn} \sqrt{Nn} = N|K| = \|x^*\|^2$, attained at $y = x^*$ — a direct support-function argument, no KKT needed for sufficiency. $\square$

Proof of (b), $n = 2, N = 1$. For $y = (y_0, y_1) \in \mathbb{C}^2$ feasible in C_1 (so $|y_m| \leq 1$ for each m , and $S_y(0) = |F(y)(0)|^2 \leq 2$ directly, hence $|F(y)(0)| \leq \sqrt{2}$),

$$\langle x^*, y \rangle_{\mathbb{R}} = (1 - t) \operatorname{Re} y(0) + t \operatorname{Re} F(y)(0) \leq (1 - t) + t\sqrt{2} = 1 + t^2 = \|x^*\|^2,$$

with equality iff $y(0) = 1$ and $F(y)(0) = \sqrt{2}$, which forces $y(1) = \sqrt{2} - 1 = t$, i.e. $y = x^*$ — a genuine separating functional exposing x^* as a vertex-like point of C_1 even though $x^* \notin \operatorname{CAZ}(2, 1)$ ($\|x^*\|^2 = 4 - 2\sqrt{2} < 2$). For the $N = 2$ doubling $((1, t), (1, t))$, the amplitude constraints are still per-block ($\operatorname{Re} y_j(0) \leq |y_j(0)| \leq 1$ for $j = 0, 1$), but the spectral constraint is *coupled* ($S_y(0) = \sum_j |F(y_j)(0)|^2 \leq 4$, not two separate per-block bounds of 2), so the argument is not a coordinatewise repeat:

$$\begin{aligned} \langle x^*, y \rangle_{\mathbb{R}} &= (1 - t) [\operatorname{Re} y_0(0) + \operatorname{Re} y_1(0)] + t [\operatorname{Re} F(y_0)(0) + \operatorname{Re} F(y_1)(0)] \\ &\leq 2(1 - t) + t\sqrt{2} \sqrt{S_y(0)} \\ &\leq 2(1 - t) + 2\sqrt{2}t = 2(1 + t^2). \end{aligned}$$

Here the first bound uses the amplitude constraints directly, while Cauchy–Schwarz gives

$$\operatorname{Re} F(y_0)(0) + \operatorname{Re} F(y_1)(0) \leq \sqrt{2} \sqrt{S_y(0)}$$

on the coupled spectral term. The last expression matches $\|x^*\|^2 = 2(1 + t^2)$ for the doubled point. Equality forces $\operatorname{Re} y_j(0) = 1$ for $j = 0, 1$ (each term of the first bracket individually at its bound, so $y_j(0) = 1$ exactly), and in the Cauchy–Schwarz step: $\operatorname{Re} F(y_0)(0) = \operatorname{Re} F(y_1)(0) = |F(y_j)(0)|$ (parallel to $(1, 1)$ and real, nonnegative) with $S_y(0) = 4$ exactly, giving $F(y_0)(0) = F(y_1)(0) = \sqrt{2}$; since $F(y_j)(0) = y_j(0) + y_j(1) = 1 + y_j(1)$, this forces $y_j(1) = \sqrt{2} - 1 = t$ for both blocks, i.e. $y = x^*$ — again a genuine separating functional, uniquely exposing the doubled point. Part (c) follows immediately from the existence of one such retained point. $\square$

Remark 5.2. Theorem A guarantees convergence to *some* fixed point; Theorem B shows the fixed-point set genuinely contains non-CAZAC points that trap every selection rule. The remaining content of “the algorithm works” is a probabilistic/basin-of-attraction statement (Q6 below), not a deterministic one.

6 Theorem C — finite exact termination at regular CAZAC limits

Call $x^* \in \operatorname{CAZ}(n, N)$ **regular** (hypothesis (R)) if the only linear dependency among the gradients of the active constraints at x^* is the one forced by Parseval’s identity; equivalently the dependency space K (Section 7 makes this precise) has $\dim K = 1$. Let $\sigma_s > 0$ denote the associated smallest singular value at a regular point.

Lemma 6.1 (Quantitative Newton-type error bound). *Let $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^q$ be a quadratic map (each component a polynomial of degree ≤ 2), $\Phi(x^*) = 0$, $A := D\Phi(x^*)$ surjective with smallest singular value $\sigma > 0$, and let M be a Lipschitz constant for $D\Phi$ (finite and global, since $D\Phi$ is affine). Then for every x with $\|x - x^*\| \leq \sigma/(8M)$ and $\|\Phi(x)\| \leq \sigma^2/(16M)$,*

$$\operatorname{dist}(x, \Phi^{-1}(0)) \leq \frac{2}{\sigma} \|\Phi(x)\|.$$

Proof. Since Φ is quadratic, Taylor's expansion is exact: $\Phi(y+v) = \Phi(y) + D\Phi(y)v + B(v, v)$ with B symmetric bilinear and $\|B(v, v)\| = \frac{1}{2}\|(D\Phi(y+v) - D\Phi(y))v\| \leq \frac{M}{2}\|v\|^2$. Run the modified Newton iteration $z_0 = x$, $z_{i+1} = z_i - A^+\Phi(z_i)$, where A^+ is the Moore–Penrose pseudoinverse ($AA^+ = I_q$ since A is surjective, $\|A^+\| = 1/\sigma$). Write $b_i = \|\Phi(z_i)\|$, $d_i = -A^+\Phi(z_i)$, so $\|d_i\| \leq b_i/\sigma$. Then

$$\Phi(z_{i+1}) = \Phi(z_i) - D\Phi(z_i)A^+\Phi(z_i) + B(d_i, d_i) = (A - D\Phi(z_i))A^+\Phi(z_i) + B(d_i, d_i),$$

so, using $\|A - D\Phi(z_i)\| \leq M\|z_i - x^*\|$,

$$b_{i+1} \leq \frac{M\|z_i - x^*\|}{\sigma} b_i + \frac{M}{2\sigma^2} b_i^2. \quad (*)$$

Claim: every z_i satisfies $\|z_i - x^*\| \leq \sigma/(4M)$ and $b_i \leq b_0 2^{-i}$. By induction: given these, $(*)$ gives $b_{i+1} \leq \frac{1}{4}b_i + \frac{1}{32}b_i < \frac{1}{2}b_i$ (using $b_i \leq b_0 \leq \sigma^2/(16M)$). Summing the geometric increments, $\sum_i \|d_i\| \leq (b_0/\sigma) \sum_i 2^{-i} = 2b_0/\sigma \leq \sigma/(8M)$, so $\|z_{i+1} - x^*\| \leq \|x - x^*\| + 2b_0/\sigma \leq \sigma/(4M)$, closing the induction. The z_i are Cauchy with limit z_∞ ; $b_i \rightarrow 0$ and continuity of Φ give $\Phi(z_\infty) = 0$, and $\|z_\infty - x\| \leq 2b_0/\sigma$. No convexity is used anywhere in this argument. $\square$

Fix $x^* \in \text{CAZ}(n, N)$ and let $H = \{c \in \mathbb{R}^n : \sum_k c_k = 0\}$, P_H the orthogonal projection onto H . Define the **balanced constraint map** $\Phi_s(x) = (g(x), P_H h(x)/n) \in \mathbb{R}^{Nn} \times H$, where $g_{j,m}(x) = |x_j(m)|^2 - 1$ and $h_k(x) = S_x(k) - Nn$ (so $g(x) = 0$ is the amplitude constraint and $h(x) = 0$ is the balanced spectral constraint).

Lemma 6.2 (Zero set). $\Phi_s^{-1}(0) = \text{CAZ}(n, N)$, globally.

Proof. If $g(x) = 0$ then x is unimodular, so by Parseval $\sum_k h_k(x) = n\|x\|^2 - nNn = 0$, i.e. $h(x) \in H$; then $P_H h(x) = 0$ forces $h(x) = 0$, so $x \in \text{CAZ}(n, N)$. Conversely $\text{CAZ}(n, N) \subseteq \{g = 0, h = 0\} \subseteq \Phi_s^{-1}(0)$. $\square$

Lemma 6.3 (Uniform derivative bounds). *The derivative map $D\Phi_s$ is globally $2\sqrt{2}$ -Lipschitz. At any $x^* \in \text{CAZ}(n, N)$ it also satisfies $\|D\Phi_s(x^*)\|_{\text{op}} \leq 2\sqrt{N+1}$. Both bounds are independent of n .*

Proof. (Lipschitz.) The rows of Dg are $2x_j(m)$, supported on disjoint real 2-planes, so $\|Dg(x) - Dg(y)\|_F = 2\|x - y\|$. The h_k -row of Dh/n in block j is (the real form of) $2\hat{x}_j(k)f_k/n$ ($f_k(m) = e^{2\pi i k m/n}$), so by Parseval $\|Dh(x)/n - Dh(y)/n\|_F^2 = 4\|x - y\|^2$; P_H is a contraction. Stacking: Lipschitz constant $\leq \sqrt{4+4} = 2\sqrt{2}$. (Operator norm at x^* .) $Dg(x^*)Dg(x^*)^\top = 4I$ (disjoint unit 2-planes, $|x_j^*(m)| = 1$); the Gram matrix of the unprojected scaled spectral rows is $4N\delta_{kk'}$ (Parseval per block, $S_{x^*} \equiv Nn$). The two blocks have operator norms 2 and $2\sqrt{N}$; the stack has norm $\leq \sqrt{4+4N} = 2\sqrt{N+1}$. $\square$

Hypothesis (R) restated exactly. $D\Phi_s(x^*)$ is surjective onto $\mathbb{R}^{Nn} \times H$ iff $\dim K = 1$, where K is the dependency space of Section 7: a pair (a, c) with $c \in H$ annihilates the row space of $D\Phi_s(x^*)$ iff $\sum_{j,m} a_{j,m} Dg_{j,m} + \sum_k (c_k/n) Dh_k = 0$, and by the definition of K these solutions are parametrized by $c \in K$ (with a determined by c). The Parseval solution $c = (1, \dots, 1) \notin H$, so the annihilator inside $\mathbb{R}^{Nn} \times H$ has dimension $\dim(K \cap H) = \dim K - 1$ (K always contains the constant vector, which is not in H); hence surjectivity $\iff \dim K = 1$, matching hypothesis (R) as stated above. Write $\sigma_s > 0$ for the smallest singular value of $D\Phi_s(x^*)$ when (R) holds, and $\rho := \sigma_s/(16\sqrt{2})$ (so $\sigma_s/(8M) = \rho$ with $M = 2\sqrt{2}$, Lemma 6.3).

Theorem 6.4 (Theorem C — Proved conditional on (R)). *Near a regular $x^* \in \text{CAZ}(n, N)$:*

- (a) Sharp (linear) growth: $D(x) \geq \frac{\sigma_s}{2\sqrt{2}} \text{dist}(x, \text{CAZ}(n, N))$ for $x \in C_1$ near x^* , where $D(x) = Nn - \|x\|^2$.
- (b) KL exponent 0: $\text{dist}(0, \partial F(x)) \geq c_0 := \sigma_s/(8\sqrt{2})$ for all non-CAZAC x in a neighborhood of x^* .
- (c) Finite exact arrival: if the Theorem A limit $\bar{x}$ is a regular CAZAC point, then $x^\ell = \bar{x}$ **exactly** for all sufficiently large ℓ . The number of “large” steps before exact arrival is at most $(Nn - \|x^0\|^2)/c_0^2$.

Proof. (a) The objective deficit is simultaneously the ℓ^1 slack of both constraint families: $D(x) = \sum_{j,m} (1 - |x_j(m)|^2) = \frac{1}{n} \sum_k (Nn - S_x(k))$, all terms ≥ 0 . In particular,

$$\begin{aligned} \|\Phi_s(x)\| &= \sqrt{\|g(x)\|^2 + \|P_H h(x)/n\|^2} \\ &\leq \sqrt{D(x)^2 + D(x)^2} = \sqrt{2}D(x). \end{aligned}$$

Case 1: $\sqrt{2}D(x) \leq \sigma_s^2/(16M)$, $M = 2\sqrt{2}$. Apply Lemma 6.1 to $\Phi = \Phi_s$ (quadratic; zero set $\text{CAZ}(n, N)$ by Lemma 6.2; surjective at x^* by (R)): $\text{dist}(x, \text{CAZ}(n, N)) \leq (2/\sigma_s)\|\Phi_s(x)\| \leq (2\sqrt{2}/\sigma_s)D(x)$, i.e. $D(x) \geq (\sigma_s/(2\sqrt{2}))\text{dist}(x, \text{CAZ}(n, N))$. *Case 2:* $\sqrt{2}D(x) > \sigma_s^2/(16M)$, i.e. $D(x) > \sigma_s^2/64$. Since x is in the stated neighborhood, $\text{dist}(x, \text{CAZ}(n, N)) \leq \|x - x^*\| \leq \rho = \sigma_s/(16\sqrt{2})$, so $(\sigma_s/(2\sqrt{2}))\text{dist}(x, \text{CAZ}(n, N)) \leq \sigma_s^2/64 < D(x)$: the bound holds directly. Either case gives (a).

(b) Let $d := \text{dist}(x, \text{CAZ}(n, N)) > 0$ and pick a nearest $z \in \text{CAZ}(n, N)$ (exists, $\text{CAZ}(n, N)$ compact). Any $w \in \partial F(x)$ has the form $w = v - x$ with $v \in N_{C_1}(x)$ (the sum rule, as in Theorem A’s proof). Since $z \in C_1$, the normal-cone inequality gives $\langle v, z - x \rangle_{\mathbb{R}} \leq 0$, hence, using $\|z\|^2 = Nn$,

$$\langle w, z - x \rangle_{\mathbb{R}} \leq -\langle x, z - x \rangle_{\mathbb{R}} = \|x\|^2 - \langle x, z \rangle_{\mathbb{R}} = -\frac{1}{2}(Nn - \|x\|^2) + \frac{1}{2}d^2 = -(F(x) - F(x^*)) + \frac{d^2}{2},$$

so Cauchy–Schwarz on the left gives $\|w\|d \geq (F(x) - F(x^*)) - d^2/2$. By (a) (in the form $F(x) - F(x^*) \geq (\sigma_s/(4\sqrt{2}))d$, i.e. (a) with $D(x) = 2(F(x) - F(x^*))$), dividing by d : $\|w\| \geq (F(x) - F(x^*))d - d/2 \geq \sigma_s/(4\sqrt{2}) - d/2$. Since $d \leq \|x - x^*\| \leq \rho = \sigma_s/(16\sqrt{2})$, this gives $\|w\| \geq \sigma_s/(4\sqrt{2}) - \sigma_s/(32\sqrt{2}) = (7/32)\sigma_s/\sqrt{2} \geq \sigma_s/(8\sqrt{2}) = c_0$. Taking the infimum over $w \in \partial F(x)$ proves (b).

(c) (*Full chain.*) Let $\bar{x}$ be the Theorem A limit, regular with constants ρ, σ_s, c_0 as in (a)/(b). Since $x^\ell \rightarrow \bar{x}$ and $\|x^{\ell+1} - x^\ell\| \rightarrow 0$ (Theorem A’s square-summability), fix L_0 so that for all $\ell \geq L_0$: $x^{\ell+1}$ lies in the (b)-neighborhood of $\bar{x}$, and $\|x^{\ell+1} - x^\ell\| < c_0$. By the exact identity from Theorem A’s proof, $w^{\ell+1} := x^\ell - x^{\ell+1} \in \partial F(x^{\ell+1})$ with $\|w^{\ell+1}\| = \|x^{\ell+1} - x^\ell\|$ *exactly*, at every ℓ – no approximation. Suppose toward contradiction $x^{\ell+1} \notin \text{CAZ}(n, N)$ for some $\ell \geq L_0$. Then $x^{\ell+1}$ is a non-CAZAC point inside the (b)-neighborhood, so (b) gives $\text{dist}(0, \partial F(x^{\ell+1})) \geq c_0$. But $w^{\ell+1} \in \partial F(x^{\ell+1})$ and $\|w^{\ell+1}\| < c_0$ by choice of L_0 — a contradiction, since $w^{\ell+1}$ itself witnesses a subgradient of norm $< c_0$. Hence $x^{\ell+1} \in \text{CAZ}(n, N)$ **exactly**: the identity is exact and (b)’s bound is a hard floor on every non-CAZAC point in the neighborhood, so a subgradient below that floor cannot occur unless the point already lies on $\text{CAZ}(n, N)$. Write $z := x^{\ell+1} \in \text{CAZ}(n, N)$. The next step maximizes $\langle z, y \rangle_{\mathbb{R}}$ over $y \in C_1$; by Cauchy–Schwarz and Prop. 3.2 ($\|z\| = \sqrt{Nn}$ is the global max-norm), $\langle z, y \rangle_{\mathbb{R}} \leq \|z\|\|y\| \leq Nn$ with equality iff $y = z$ — the argmax is the singleton $\{z\}$, so $x^{\ell+2} = z$, and by induction $x^m = z$ for all $m > \ell$; in particular $\bar{x} = z \in \text{CAZ}(n, N)$. The step-count bound is Theorem A’s $\sum_\ell \|x^{\ell+1} - x^\ell\|^2 \leq Nn - \|x^0\|^2$ telescoped against c_0^2 per large step (increment $\geq c_0$); since every step after arrival has increment exactly 0 (the sequence is constant at z), every large step occurs strictly before arrival, so this global count already *is* the count of large steps before arrival, with no separate argument needed to exclude post-arrival steps. $\square$

Remark 6.5. This is a genuine proof by contradiction, not a heuristic squeeze: an exact identity (equality, from Theorem A) is jointly satisfiable with a hard lower bound (inequality, from (b)) on the same quantity only at $\text{dist} = 0$. Parts (a), (b), (c) share one status tag because they rest on exactly the same two inputs — hypothesis (R) at $\bar{x}$ (giving σ_s, ρ) and the exact H2 identity from Theorem A’s proof — and (c) adds no assumption beyond the theorem’s own hypothesis “ $\bar{x} \in \text{CAZ}(n, N)$ ” plus the unconditional singleton-argmax fact.

Remark 6.6 (The $N = 1$ vs. $N = 2$ gap is local-rate-invariant). Theorem C’s constants do not distinguish $N = 1$ from $N = 2$: first-order sharpness at a regular point decides the local geometry identically. The empirically observed difference in success probability between the single-sequence and coupled formulations is therefore a global (basin-of-attraction) phenomenon, not a local-rate one.

Remark 6.7 (Novelty check against generic identifiability theory). Neither the Lewis–Luke–Malick theory of local linear convergence for averaged/alternating projections [6] nor the Drusvyatskiy–Lewis generic identifiability program [7] delivers Theorem C. The former’s core hypothesis, strongly regular (transversal) intersection, is exactly what Section 7 shows *fails* at every CAZAC point (the universal Parseval dependency), and even where satisfiable its conclusion is R-linear convergence of a different algorithm, never finite exact arrival. The latter’s genericity condition is plausibly satisfied at CAZAC points, but its machinery yields only a qualitative locally-minimal-identifiable-set statement, not a quantitative sharp-growth constant, KL exponent, or finite-step termination guarantee.

7 Theorem D and regularity (R)

Write $x_u(m) = \omega^{-uT_m}$ ($\omega = e^{2\pi i/n}$, $T_m = m(m+1)/2$, n odd) or $x_u(m) = \omega_{2n}^{-um^2}$ ($\omega_{2n} = e^{i\pi/n}$, n even) for the Zadoff–Chu sequence of root u , $\gcd(u, n) = 1$ [8]. At a CAZAC tuple all constraints are active; write

$$K = \{c \in \mathbb{R}^n : \overline{x_j(m)} (F^*(c \odot \hat{x}_j))(m) \in \mathbb{R} \ \forall j, m\}$$

for the active-gradient dependency space (the tangent space of the constraint intersection has dimension $(N-1)n + \dim K$; hypothesis (R) is $\dim K = 1$).

Theorem 7.1 (Theorem D — Proved for all $n \geq 1$, odd and even). *Let $n = \prod_p p^{a_p}$, $d_{\max} = \prod_p p^{\lfloor a_p/2 \rfloor}$ (the largest d with $d^2 \mid n$). Then for every $n \geq 1$, every $N \geq 1$, every tuple of coprime roots $u_1, \dots, u_N$,*

$$K = \{c \in \mathbb{R}^n : c(k + d_{\max}) = c(k) \ \forall k\}, \quad \dim_{\mathbb{R}} K = d_{\max},$$

independent of N , of the roots, and of the parity of n .

Proof. Step 0. A dependency $\sum_{j,m} a_{j,m} \nabla g_{j,m} + \sum_k c_k \nabla h_k = 0$ evaluated on a perturbation is equivalent to $F^*(c \odot \hat{x}_j)$ being coordinatewise a real multiple of x_j — the defining condition of K ; dependencies with $c = 0$ force $a = 0$ since amplitude gradients have disjoint supports. So the dependency space is (isomorphic to) K .

Step 1 (self-transform). For odd n , writing $k = us$, $\hat{x}_u(k) = G_u \omega^{uT_{u^{-1}k}}$ with $G_u = \sum_j \omega^{-uT_j}$, $|G_u|^2 = n$ by Parseval, using the exact identity $T_{m+s} = T_m + T_s + ms$. For even n , the parallel identity holds with T_m replaced by m^2 , exponents tracked mod $2n$: $\hat{x}_u(k) = G_u \omega_{2n}^{us^2}$, using $\omega_{2n}^{v(j+n)^2} = \omega_{2n}^{vj^2}$ (an exact integer identity valid because n is even).

Step 2 (reduction to a correlation equation). Fix a block with root u and write $q_m = \overline{x_u(m)} (F^*(c \odot \hat{x}_u))(m)$. Substituting Step 1, reindexing $k = us \bmod n$ (a bijection; $c'_s := c_{us}$ again real), and using the same addition-law identity as Step 1 gives, in both parities, $q_m = G_u \sum_s c'_s \phi(s+m)$, where $\phi(j) = \omega^{uT_j}$ (odd n) or $\phi(j) = \omega_{2n}^{uj^2}$ (even n). Since c' is real, $q_m \in \mathbb{R}$ for all m iff $q_m - \overline{q_m} = 0$:

$$\sum_s c'_s e(s+m) = 0 \quad \forall m, \quad e(j) := G_u \phi(j) - \overline{G_u \phi(j)}$$

— the vanishing of a convolution against the *reality-defect* kernel e , not mere reality of $\sum_s c'_s \phi(s+m)$ itself, which would be a strictly stronger, false requirement: ϕ alone is not conjugate-symmetric, and $\hat{\phi}$ never vanishes (it is a pure Gauss sum of constant modulus $\sqrt{n}$), so that reading would force $\hat{c}' \equiv 0$, contradicting $\dim K \geq 1$.

Step 3 (Fourier solve). Taking the length- n DFT in m turns the convolution into $\hat{e}(t) \hat{c}'(-t) = 0$ for every t , i.e. $\hat{c}'(t) = 0$ wherever $\hat{e}(-t) \neq 0$. Writing $\sigma = u^{-1}t$ and using Step 1's formula for ϕ with $|G_u|^2 = n$: $\hat{e}(t) = n(\omega^{u(\sigma-T_\sigma)} - \omega^{uT_\sigma})$ (odd n , using $T_{-\sigma} = T_\sigma - \sigma$) or $\hat{e}(t) = n(\omega_{2n}^{-u\sigma^2} - \omega_{2n}^{u\sigma^2})$ (even n). In both parities $\hat{e}(t) = 0$ exactly when $\sigma^2 \equiv 0 \pmod{n}$ (odd: divide the exponent congruence by u to get $\sigma - T_\sigma \equiv T_\sigma$, i.e. $\sigma \equiv 2T_\sigma = \sigma^2 + \sigma$; even: $2u\sigma^2 \equiv 0 \pmod{2n} \iff u\sigma^2 \equiv 0 \pmod{n}$, and u is a unit): the zero set is the subgroup $Z_0 = \{t : t^2 \equiv 0 \pmod{n}\} = m_0 \mathbb{Z}_n$, $|Z_0| = n/m_0 = d_{\max}$ where $m_0 = \prod_p p^{\lceil a_p/2 \rceil}$.

Step 4. $\hat{c}'(t) = \hat{c}(u^{-1}t)$, so “ $\text{supp } \hat{c}' \subseteq Z_0$ ” is the same condition for every root and block, giving $K = \{c : \text{supp } \hat{c} \subseteq m_0 \mathbb{Z}_n\}$, i.e. the $d_{\max}$ -periodic real vectors, of real dimension $d_{\max}$. This count is read off directly from $|Z_0| = d_{\max}$ and is insensitive to parity: an alternate derivation via counting conjugate-symmetric pairs $\{t, -t \bmod n\}$ in Z_0 would need to treat the self-paired frequency $t = n/2$ separately when $n \equiv 0 \pmod{4}$ places $n/2 \in Z_0$, but that bookkeeping is specific to the pair-counting route and does not arise here, since Steps 1–4 characterize K by its Fourier support directly. $\square$

Remark 7.2 (Scope of the written derivation). Steps 2–3 evaluate the same kind of quadratic exponential sum as Step 1's self-transform identity, via the same completion-of-square technique. The definition of the reality-defect kernel e is the one place this computation is easy to get wrong: dropping the conjugate term mis-simplifies the zero set to the wrong condition, structurally the same error as transferring χ 's self-duality naively (i.e. without the affine correction) to Björck's construction below (Lemma 7.8's own point of departure). The *conclusion* $\dim K = d_{\max}$ is independently corroborated: by hand at $n = 4$ (self-paired case) and $n = 9$ (odd, non-self-paired) via direct enumeration of Z_0 , and numerically across all $21 + 43 = 64$ tested Zadoff–Chu configurations (both parities; Section 7's corollaries below cite the exact scripts). No counterexample or numerical discrepancy has been found at any tested configuration.

Corollary 7.3. *Hypothesis (R) holds at every Zadoff–Chu tuple iff n is squarefree — in particular at every prime n , odd or $n = 2$. Theorem C is therefore **unconditional** whenever the Theorem A limit lies on the symmetry orbit of a Zadoff–Chu tuple and n is squarefree.*

Corollary 7.4. *At Zadoff–Chu tuples with squarefree n (either parity), the CAZAC tangent dimension is exactly $(N - 1)n + 1$: a 1-dimensional phase circle for $N = 1$ versus an $(n + 1)$ -dimensional manifold for $N = 2$.*

Numerical verification uses `experiments/verify_zc_regularity.py` for odd lengths

$$n \in \{3, 5, 7, 9, 15, 21, 25, 27, 45, 49, 75\}$$

(21 configurations), and `experiments/verify_even_n_regularity.py` for the even lengths

$$n \in \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 24, 36, 50\}$$

(43 configurations). All match $\dim K = d_{\max}$ exactly. No Chebotarëv-type nonvanishing-minors argument is needed: the DFT of a chirp is a chirp, collapsing the dependency analysis to counting solutions of $t^2 \equiv 0 \pmod n$. Both scripts are in the companion repository [16].

Remark 7.5 (Prior art bearing on the lower bound). Three older, independent constructions already anticipate part of Theorem D — its *existence* half, that $\dim K$ can be as large as $d_{\max}$, not just 1 — via routes substantially different from, and more elementary than, this paper’s own. **Backelin (1989)** [10]: for $n = m^2\ell$ with ℓ squarefree, an explicit algebraic construction gives an $(m - 1)$ -parameter family of solutions to the cyclic- n -roots system (the polynomial system underlying CAZAC/ biunimodular sequences), which at $m = d_{\max}$, restricted to unimodular parameters, matches $d_{\max} - 1$ free real phases — the same count as $\dim K$ once the always-present global-phase direction is added back. **Popović (1992)** [11]: modulating a Zadoff–Chu sequence of length $n = sm^2$ by an *arbitrary* unimodular sequence of period m again satisfies the ideal periodic autocorrelation property (their Theorem 1); at $m = d_{\max}$ this gives an explicit $d_{\max}$ -real-parameter family of CAZAC points *through every Zadoff–Chu tuple* — confirmed numerically that this family exists exactly as claimed and has the predicted dimension at $n = 4$ ($d_{\max} = 2$: an arbitrary period-2 phase pair gives a verified 2-parameter family of exact CAZAC points; script: `experiments/verify_popovic_n4.py` in the companion repository [16]). **Björck–Saffari (1995)** [13]: a further, more general construction (any constant-amplitude length- N seed sequence and any permutation, not just a phase modulation of a fixed Zadoff–Chu sequence) gives explicit continuous families of CAZAC sequences of length N^2 , extended by a further construction (with an initial parity condition removed by a Kronecker-product trick) to every non-squarefree length unconditionally. Of the three, Popović’s is the most directly comparable to this paper’s setup (a family literally through a Zadoff–Chu tuple), and we have **not** worked out the precise identification between any of these three explicit families and this paper’s dependency space K (a natural next step; the dimension matches above are evidence for, not proof of, such an identification). What Theorem D adds beyond all three: the *exact* count (no more than $d_{\max}$, not merely at least $d_{\max}$), at every Zadoff–Chu tuple and both parities; the reusable chirp-DFT-closure mechanism itself, which extends to Björck’s non-Zadoff–Chu construction and to ruling out cubic-or-higher phase; and the connection to hypothesis (R) and Theorem C’s convergence rate, none of which the 1989–1995 constructions address. Separately: Björck and Saffari also *conjectured* the complementary finiteness statement — that at squarefree n the full CAZAC variety (not just the Zadoff–Chu orbit) is finite up to phase — which remains open in general (settled only for prime n , by Haagerup [15]); this is the natural generalization of exactly the gap Q4 below leaves open (regularity at non-Zadoff–Chu CAZAC points), not a new question invented by this paper.

7.1 The chirp-argument schema, and where it does and does not generalize

Stripped of Zadoff–Chu-specific notation, Theorem D’s proof needs exactly: (1) a closure property, $\widehat{x_u} = G_u \cdot x_u \circ \phi_u$ for a scalar G_u and an index-set automorphism ϕ_u ; (2) an algebraic addition law for the phase making the key substitution exact; (3) applying the closure twice, once to reduce the dependency condition to a correlation equation, once more to diagonalize that equation into a congruence on the Fourier support of the kernel. This replaces a linear-algebra rank question with a congruence-counting question, and is a genuinely reusable schema, not a one-off computation.

Proposition 7.6 (Positive lead — Proved classically, newly connected here). *The Legendre symbol χ modulo an odd prime p satisfies $\hat{\chi}(k) = \chi(k) g(\chi)$ for $k \neq 0$ ($g(\chi)$ the Gauss sum, $|g(\chi)|^2 = p$): χ is, up to a scalar, its own DFT, the same closure shape used by Theorem D, with trivial reindexing.*

Proposition 7.7 (Obstruction to this specific route — not a general impossibility result). *For cubic*

and higher-degree phase (“algebraic chirp”) sequences, Theorem D’s specific closure mechanism — completing the phase by a shift, the way Step 1 completes the square — fails, for the same reason completing the square is special to degree 2: the addition law $(m+s)^d = m^d + ds m^{d-1} + \binom{d}{2} s^2 m^{d-2} + \dots + s^d$ has a mixing term $ds m^{d-1}$ of degree $d-1$ in m , which is linear in m (absorbable into a pure reindexing of the phase, exactly as Step 1 does) only when $d=2$; for $d \geq 3$ and $\gcd(d, n) = 1$, matching the cross term $-km$ (linear in m) forces $ds \equiv 0 \pmod{n}$, hence $s \equiv 0$, so no nontrivial shift completes the d -th power the way Step 1 completes the square. Separately, the specific quadratic Gauss sum G_u entering Theorem D’s Step 1 has magnitude exactly $\sqrt{n}$, forced by Parseval applied to the Zadoff–Chu sequence itself (not by a general closed-form evaluation: classical quadratic Gauss sums have magnitude $\sqrt{n}$ for n odd, 0 for $n \equiv 2 \pmod{4}$, and $\sqrt{2n}$ for $n \equiv 0 \pmod{4}$); Weyl sums of degree ≥ 3 have no comparable exact evaluation, only upper bounds, so even if some other closure map existed it would not obviously carry the exact-modulus property Step 1 needs. **Together these show the specific self-transform mechanism used in Theorem D’s proof does not extend to cubic or higher phase**; this is an obstruction to this route, not a proof that no route exists, and the mixing-term argument itself is stated only for $\gcd(d, n) = 1$ — when d and n share a factor, $ds \equiv 0 \pmod{n}$ can have nontrivial solutions and the argument as given does not cover that case.

7.2 The Björck case: a partial result, not a proof

Björck’s construction (odd prime p , Legendre symbol χ) [12] defines $b(m) = e^{i\theta(m)}$ with θ a two-valued function of $\chi(m)$, differing by $p \pmod{4}$. Unlike χ itself, b is not literally self-dual under the DFT, but a different closure fact rescues part of the argument.

Lemma 7.8 (Proved, exact symbolic computation). *b is exactly **affine** in χ : $b = A \cdot \mathbf{1} + B \cdot \chi + C \cdot \delta_0$ with $C = 1 - A$, where*

$$\begin{aligned} p \equiv 1 \pmod{4}: \quad & A = \cos \alpha, \quad B = i \sin \alpha, \\ & \alpha = \arccos \frac{1}{1 + \sqrt{p}}; \\ p \equiv 3 \pmod{4}: \quad & A = \frac{1 + e^{i\beta}}{2}, \quad B = \frac{1 - e^{i\beta}}{2} = C, \\ & \beta = \arccos \frac{1 - p}{1 + p}. \end{aligned}$$

The 3-dimensional space $T := \text{span}_{\mathbb{C}}\{\mathbf{1}, \chi, \delta_0\}$ closes under the DFT: $\hat{b} = C \cdot \mathbf{1} + Bg \cdot \chi + Ap \cdot \delta_0$ (using $F(\mathbf{1}) = p\delta_0$, $F(\chi) = g\chi$, $F(\delta_0) = \mathbf{1}$, $g = g(\chi)$ the Gauss sum, $g = \sqrt{p}$ for $p \equiv 1 \pmod{4}$ and $g = -i\sqrt{p}$ for $p \equiv 3 \pmod{4}$, with the sign fixed by this paper’s DFT convention $\hat{x}(k) = \sum_m x(m)\omega^{-km}$, i.e. $g = \overline{G(\chi)}$ for the classical Gauss sum $G(\chi) = \sum_m \chi(m)\omega^m = i\sqrt{p}$ at $p \equiv 3 \pmod{4}$).

Lemma 7.9 (Proved). *Let G be the group of quadratic-residue units mod p (order $q = (p-1)/2$), acting by decimation. $b, \hat{b}$ are pointwise G -invariant, so K is a G -invariant subspace of $\mathbb{R}^p$. $\mathbb{R}^p \cong \mathbf{1}_{\{0\}} \oplus \mathbb{R}[G] \oplus \mathbb{R}[G]$ as a G -module, with trivial-isotypic piece exactly $T \cap \mathbb{R}^p$ (dimension 3).*

Theorem 7.10 (Proved, both residue classes). *$K \cap T = \text{span}\{\mathbf{1}\}$; i.e. restricted to the 3-dimensional symmetry-invariant candidate space, the only dependency is the Parseval one, for every prime p .*

Proof. Restricting the dependency condition $\overline{b(m)} (F^*(c \odot \hat{b}))(m) \in \mathbb{R}$ for all m to $c = \alpha \mathbf{1} + \beta \chi + \gamma \delta_0 \in T$ collapses — since $b, \hat{b} \in T$ closes under pointwise product and F — to exactly three scalar equations, one evaluation each at $m = 0$, $m \in \text{QR}$, $m \in \text{QNR}$ (the condition depends on m

only through $\chi(m) \in \{0, 1, -1\}$; α never appears in any of the three (consistent with $\mathbf{1} \in K$ unconditionally), so only the β, γ coefficients are at stake. Use $\alpha_0 := \alpha$ and $\beta_0 := \beta$, the same angles fixed in Lemma 7.8 (not a phase of Bg : for $p \equiv 1 \pmod{4}$, $Bg = i \sin \alpha \cdot \sqrt{p}$ is purely imaginary, never real, so no polar-decomposition phase of Bg itself is what is meant here).

For $p \equiv 1 \pmod{4}$, eliminating the $m = 0$ equation (which involves only β) leaves:

$$s\sqrt{p}\beta = 0, \quad -s\gamma(1 + c(p-1)) = 0 \quad (c = \cos \alpha_0 = \frac{1}{1+\sqrt{p}}, s = \sin \alpha_0).$$

$\alpha_0 = \arccos \frac{1}{1+\sqrt{p}} \in (0, \pi/2)$ since $\frac{1}{1+\sqrt{p}} \in (0, 1)$, so $s = \sin \alpha_0 > 0$; $s\sqrt{p} \neq 0$ forces $\beta = 0$ directly. Also $c = \cos \alpha_0 = \frac{1}{1+\sqrt{p}} \in (0, 1)$ exactly (not merely bounded, and not ± 1), so $1 + c(p-1) > 1 > 0$ for every $p \geq 5$ (both terms of $c(p-1)$ positive), hence $1 + c(p-1) \neq 0$ forces $\gamma = 0$.

For $p \equiv 3 \pmod{4}$, the QR and QNR equations (the $m = 0$ equation is linearly dependent on these two) are:

$$\sqrt{p}(1 - \cos \beta_0) \beta + \frac{p-1}{2} \sin \beta_0 \gamma = 0, \quad \frac{p+1}{2} \sin \beta_0 \gamma = 0.$$

$\beta_0 = \arccos \frac{1-p}{1+p}$ with $\frac{1-p}{1+p} \in (-1, 1)$ strictly for every $p \geq 3$, so $\beta_0 \in (0, \pi)$ and $\sin \beta_0 > 0$; the second equation then forces $\gamma = 0$ directly, since $\frac{p+1}{2} \sin \beta_0 \neq 0$; substituting into the first, $\sqrt{p}(1 - \cos \beta_0)\beta = 0$ forces $\beta = 0$ as well, since $1 - \cos \beta_0 \neq 0$ ($\beta_0 \neq 0$).

In both residue classes the system forces $\beta = \gamma = 0$ with α free, giving $K \cap T = \text{span}\{\mathbf{1}\}$ exactly. $\square$

Remark 7.11 (What is proved and what remains open). Lemma 7.8 and Lemma 7.9 and Theorem 7.10 are complete proofs. What remains **unproved** is whether K meets any of the $q-1$ nontrivial isotypic pieces of $\mathbb{R}^p$ under G : by Schur's lemma the dependency operator restricted to each such (complex 2-dimensional) block is a 2×2 matrix, structurally a Jacobi-sum-type bilinear expression, and its nonvanishing was not proved. For $p \equiv 1 \pmod{4}$, an exact closed form for the block (in terms of twisted Gaussian periods $\rho_R(k), \rho_N(k)$ and sequence-specific factors u, v, w, p_f) gives $\det = -\frac{1}{4}[e(-k/q)\rho_R(k)^2 u(k)v(k) - \rho_N(k)^2 p_f(k)w(k)]$; $\rho_R(k) = \frac{1}{2}[g(\lambda_k) + g(\lambda_k \chi)]$, $\rho_N(k) = e(-k/2q)\frac{1}{2}[g(\lambda_k) - g(\lambda_k \chi)]$ for λ_k the order- $(p-1)$ character with $\lambda_k(g_0) = e(k/(p-1))$ (g_0 a primitive root mod p). **One route to $\det = 0$ is now ruled out unconditionally**: the route where $\rho_R(k)$ and $\rho_N(k)$ vanish simultaneously (which would zero out the entire 2×2 block, hence $\det$). $\rho_R(k) = 0$ or $\rho_N(k) = 0$ individually requires $g(\lambda_k) = \pm g(\lambda_k \chi)$, and the classical Gauss–Jacobi identity $g(\psi)g(\chi) = J(\psi, \chi)g(\psi\chi)$ re-expresses this as $J(\lambda_k, \chi) = \pm g(\chi)$; since $J(\lambda_k, \chi) \in \mathbb{Q}(\zeta_{p-1})$ (a $\mathbb{Z}$ -linear combination of $(p-1)$ -th roots of unity) while $g(\chi) \in \mathbb{Q}(\zeta_p) \setminus \mathbb{Q}$ (classically $\pm\sqrt{p}$ or $\pm i\sqrt{p}$, never rational) and $\mathbb{Q}(\zeta_{p-1}) \cap \mathbb{Q}(\zeta_p) = \mathbb{Q}$ (since $\gcd(p-1, p) = 1$), the equality $J(\lambda_k, \chi) = \pm g(\chi)$ would force $g(\chi) \in \mathbb{Q}$, a contradiction. So $\rho_R(k), \rho_N(k) \neq 0$ for every prime and every nontrivial k , **proved**, not merely checked (script: `experiments/verify_bjorck_jacobi_unification.py` in the companion repository [16]) — a clean standalone fact, but **one that does not, by itself, narrow the determinant question**: neither $\rho_R(k) = 0$ nor $\rho_N(k) = 0$ was ever a necessary or a sufficient condition for $\det = 0$ on its own (e.g. at $\rho_R(k) = 0$, $\det$ collapses to $\frac{1}{4}\rho_N(k)^2 p_f(k)w(k)$, itself zero only if p_f or w also vanishes), so this proof does not directly rule out $\det = 0$ in general. $\det$ is a *difference* of two terms, and the actual open condition is the cross-term coincidence $e(-k/q)\rho_R(k)^2 u(k)v(k) = \rho_N(k)^2 p_f(k)w(k)$, which can hold even with $\rho_R, \rho_N, u, v, w, p_f$ all individually nonzero. $u(k)v(k) - w(k)p_f(k)$ has an explicit nonzero closed form ($-16A\varepsilon g C \sin^2 \alpha$ for the constants A, C of Lemma 7.8 and $\varepsilon := (-1)^k$; itself never zero, since $A = \cos \alpha \in (0, 1)$, $C = 1 - A \neq 0$, $g = \sqrt{p} \neq 0$, and $\sin \alpha \neq 0$), confirming the cross-term is a genuinely separate condition, not reducible to ρ_R, ρ_N vanishing. Checked directly, not assumed away, **for $p \equiv 1 \pmod{4}$ in [5, 60]**: at the k minimizing $|\det|/p^2$ for each such prime, $\min(|\rho_R(k)|, |\rho_N(k)|)/\sqrt{p}$

stays in $[0.11, 0.6]$ — **not small** — even as $|\det|/p^2$ itself shrinks by three orders of magnitude, confirming that in *this* range the near-degeneracy traces to the cross-term, not to ρ_R or ρ_N (script: `experiments/verify_bjorck_crossterm_gap.py`, same repository). **This does not extend to $p = 71$:** that prime is $\equiv 3 \pmod{4}$, outside both the closed form above and this sweep’s range, and at $p = 71, k = 17$ specifically $\min(|\rho_R|, |\rho_N|)/\sqrt{p} \approx 5.3 \times 10^{-7}$ — there, unlike the $p \equiv 1 \pmod{4}$ range just checked, the near-degeneracy traces directly to ρ_R or ρ_N being small, not to the cross-term; the two residue classes are not shown to behave the same way, and the $p \equiv 3 \pmod{4}$ closed form needed to even pose the cross-term question there has not been derived. The cross-term coincidence remains open and unresolved for $p \equiv 1 \pmod{4}$. Separately, the relevant determinants are confirmed numerically nonzero at every one of 44 tested primes $p \in [5, 200]$ (both residue classes, $\dim K = 1$ throughout) — at $p = 71$ the gap is a factor of $\approx 4 \times 10^3$ smaller than at the next-nearest tested prime, a genuine near-degeneracy, independently confirmed not to be a floating-point artifact by a 60-digit-precision (mpmath) recomputation of this specific block (top singular value matches to $\sim 10^{-15}$ relative precision, the near-zero singular value moves by only $\sim 1\%$ of its own scale between float64 and 60-digit precision, not by orders of magnitude). Script: `experiments/verify_bjorck_isotypic.py` in the companion repository [16] (the 44-prime determinant check; a separately-run set of 14 primes, all contained in this same tested range, is used for the affine-closure check of Lemma 7.8, in `experiments/verify_bjorck_regularity.py`). **The overall regularity claim at Björck points is not proved in this paper:** what is established here is a full proof restricted to the symmetry-invariant subspace, an unconditional proof ruling out one route (both twisted periods vanishing at once) by which the remaining determinant could fail, and strong (but non-uniform) numerical evidence, not a proof, for the complementary directions as a whole. Full regularity — for every prime and both residue classes — is proved in the literature by an independent argument of the same general character (decimation-group isotypic decomposition, a Gauss-sum ratio shown never to be a root of unity via Stickelberger’s theorem): Benoist [14], Prop. A.3. This paper’s own contribution regarding Björck’s construction is the connection to hypothesis (R) and Theorem C’s convergence rate, not the underlying regularity fact itself.

8 Open problems

Q1 (settled). Does x^ℓ converge to a single point? Yes — Theorem 4.1.

Q2 (open, reframed). Characterize all fixed points of $(\star)$ for general n, N . The stationarity system is a KKT eigenvalue equation $x^* = (\Lambda + F^*MF)x^*$ for diagonal multipliers, structurally reminiscent of discrete prolate/Slepian time-and-band-limiting operators. Complete for $n = 2$ (four symmetry classes); open in general. A sharper sub-question: are non-CAZAC fixed points always measure-zero repellers, or can they attract an open set of initializations?

Q3 (partially answered). Why does $N = 2$ empirically always succeed where $N = 1$ sometimes fails? Corollary 7.4’s exact dimension count, $\dim \text{CAZ}(n, N) = (N - 1)n + 1$ at squarefree Zadoff–Chu tuples, suggests a mechanism (more coupled sequences means more escape directions), though this is proved only at those specific points, not as a general lower bound at every CAZAC point (which would additionally require $\text{CAZ}(n, N)$ to be a manifold there, not established in general); the pathological fixed-point classes are the same for $N = 1, 2$, so the gap is about basin measure, not existence, and remains open.

Q4 (essentially settled locally; one remaining gap). Rate of convergence near $\text{CAZ}(n, N)$: settled by Theorem C at regular points, and Theorem D now makes regularity unconditional at Zadoff–Chu points of either parity for squarefree n . Remaining: (R) at non-Zadoff–Chu CAZAC points in general is open in this paper (Björck’s own construction is resolved in the literature, [14],

though not by this paper’s methods); n -uniformity of the true sharp-growth constant is conjectural (the proved bound scales like c_N/n ; whether the true constant does too is open and not backed here by a specific numerical check). This remaining gap is not a new question: it is the natural generalization of Björck and Saffari’s own still-open finiteness conjecture (§7, remark after Corollary 7.4) — settled only at prime n , by Haagerup [15].

Q5 (open, deprioritized). What, if anything, do the preliminary solves buy? The reference implementation solves successively at $\tau = 1, 1/2, \dots, 1/10$ before resetting to $\tau = 1$. Their effect has not been tested by ablation, and no theoretical benefit is claimed.

Q6 (the central open question). For $N \geq 2$ and Gaussian initialization, does the iteration converge to $\text{CAZ}(n, N)$ with probability 1 (or $\geq 1 - c^n$)? Requires a full fixed-point census (Q2) and a basin/measure argument. No tracked experiment in this repository currently estimates that success probability for $(\star)$. The 15/50 success count at $n = 167$ printed by the tracked projection script belongs to a different IPUC-style baseline and is not evidence for this question.

Q7 (partially settled by citation, residual part open). Does every trajectory of the elliptope iteration converge to a rank-1 point? Theorem 4.7 proves single-point convergence unconditionally but not convergence to an extreme point specifically. The *pointwise* half of this question is already settled — Remark 4.9 cites [3]’s own Theorem 20: rank-1 points are exactly the fixed points whose full neighborhood converges to them, and every other fixed point admits an explicit escaping direction. What remains open is the *measure-theoretic* refinement: does the set of initializations whose trajectory converges to a non-vertex fixed point have Gaussian measure zero? [3]’s own escaping-curve argument (Prop. 19, Remark 4.9) only produces one escaping point per neighborhood, not universal escape, so it leaves open whether non-vertex elliptope points are repelling outright or could instead be collectively (not individually) attracting, the way [3]’s unrelated cone example (§3, Example 5) shows can happen in general for this style of iteration. This residual question is structurally the same conjecture as Q6 in a second, unrelated domain: if the measure is zero, it is independent evidence for Q6’s general shape; if not, CAZAC’s extra structure (the dimension count of Corollary 7.4) may be doing more work than currently credited. The numerical evidence (Prop. 4.6, 140 trials across $n \in \{3, \dots, 20\}$) shows zero exceptions, consistent with — but not a proof of — the measure-zero reading. *Attack plan:* a basin/measure argument analogous to Q6’s, using the pointwise dichotomy of Theorem 20 as the starting classification of which points can even be candidates for a positive-measure limit.

9 Verification status and provenance

Tiers. *Unconditional:* complete proof, no open hypothesis. *Proved conditional on (R):* complete proof, resting on regularity hypothesis (R). *Proved conditional on (R) and an open sub-case:* complete proof modulo one explicitly identified, unresolved lemma. *Verified numerically:* no proof; a script reproduces the claim to a stated tolerance. *Conjecture:* stated as a target, not established either way.

Principal claims.

- Propositions 3.1–3.4 (generalized power method, max-norm = CAZAC, monotone ascent, limit points fixed): *Unconditional*.
- Theorem A (global single-point convergence): *Unconditional* as a mathematical claim, but its descent-argument mechanism is a special case of Aktaş–Kroer’s [4] general Frank-Wolfe/KL framework (2025); the paper’s own contribution here is the CAZAC-to-max $\|x\|^2$ reduction

(Prop. 3.2) that makes that template apply, not the descent argument itself — see the note at Theorem A.

- Elliptope semialgebraicity and H1–H3 transfer (Props. 4.3–4.4): *Unconditional*, every n . Fixed-point continuum (Prop. 4.5): *Unconditional*, $n > 3$ *only* (infinitude by citation to [3], positive-dimensionality by semialgebraicity, plus a machine-exact illustrative identity at I_n); at $n \leq 3$ the fixed-point set is finite (per [3]’s own count), so the statement is restricted to $n > 3$, not vacuously true for all n . Single-point convergence (Theorem 4.7): *Unconditional for every n* — this is a proved theorem, not a numerical finding; the “despite an infinite continuum” framing is specific to $n > 3$, but convergence itself does not depend on that framing. The pointwise vertex/non-vertex attractive dichotomy (Remark 4.9): *Unconditional*, by direct citation to [3] Theorem 20 (not a new argument here). *Separately*, that generic trajectories escape onto a rank-1 point specifically (Prop. 4.6): *Verified numerically only* (140 trials, $n \in \{3, \dots, 20\}$) — this is not proved. The residual, measure-zero form of universal rank-1 convergence (Q7): *Conjecture*.
- Theorem B (obstruction, exposed non-CAZAC fixed point): *Unconditional* for every N (part (a)’s statement and proof cover general N : the coupled spectral condition $S_x(k) = Nn$ on K is pinned, individual per-sequence moduli are not, and only $N = 1$ collapses to a per-block modulus condition).
- Theorem C, parts (a),(b),(c), including the full contradiction-based proof of part (c): *Proved conditional on hypothesis (R)*; all three parts carry the same status tag.
- Theorem D (regularity of Zadoff–Chu points, $\dim K = d_{\max}$), **both odd and even n** : *Unconditional*, a complete proof for all n regardless of parity, with Steps 2–3 spelled out via the reality-defect kernel e (see the “Scope of the written derivation” remark after the proof) — independently corroborated by hand at $n = 4, 9$ and numerically at all 64 tested configurations. Prior art disclosed (remark after Corollary 7.4): three older, independent constructions — Backelin (1989) [10], Popović (1992) [11], Björck–Saffari (1995) [13] — each give, by unrelated elementary methods, explicit families matching the *existence* half of this count ($\dim K$ can reach $d_{\max}$) at non-squarefree n ; Theorem D’s own contribution is the exact count and the reusable mechanism, not the existence of degeneracy per se. Corollaries 7.3–7.4: *Unconditional* (Corollary 1 conditionally activates Theorem C, but is itself unconditional as a dimension/regularity statement).
- Legendre-symbol closure positive lead: classical fact, newly connected to the schema, *Unconditional* as stated (no claim of Björck regularity is made from it alone). Cubic/higher-phase obstruction: *proved for the specific completing-the-square mechanism* used in Theorem D’s proof, when $\gcd(d, n) = 1$ (the mixing term becomes nonlinear in m for $d \geq 3$); **not** a general impossibility result, and not addressed when $\gcd(d, n) > 1$.
- Björck partial result: Lemma 7.8, Lemma 7.9, Theorem 7.10 ($K \cap T = \text{span}\{\mathbf{1}\}$): *Unconditional*. Within the residual (non-symmetry-invariant) directions: $\rho_R(k), \rho_N(k) \neq 0$ for every prime and every nontrivial k is *Unconditional* (proved, an elementary cyclotomic-field argument, not merely checked — §7, Björck subsection), but neither $\rho_R(k) = 0$ nor $\rho_N(k) = 0$ was ever a necessary or a sufficient condition for the full isotypic-block determinant to vanish on its own — this rules out only the sub-case where both vanish at once, not the actual cross-term condition, which this paper’s own methods do not resolve (verified numerically only, at 44 primes, $p \in [5, 200)$). Full regularity (R) at Björck points is **not proved in this paper**; it is

available in the literature [14], via a similar Gauss-sum technique, for every prime and both residue classes.

- Q1–Q7: precise open-problem statements as in Section 8; Q1 is settled (restated as a theorem above), the rest range from “partially answered” to fully open conjectures, as stated there.

Provenance. This document was written with substantial assistance from large language models (Claude, Anthropic) under the direction of Jeffery Kline, drawing on a longer working record of derivations kept in the companion repository [16]’s `theory/` directory. Numerical claims name their supporting scripts in that repository’s `experiments/` directory. Claims whose supporting companion artifact is unavailable are identified as such in the repository’s release documentation.

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